MATH 317 Project 3: Classification of tree species from images of leaves

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I. Problem Description

We are tasked with trying to determine the species of a tree using images of the tree's leaves via a classification algorithm. This requires us to deal with a very large data matrix of the leaf images which imposes some challenges as to the kinds of operations we can do. We addressed the problem using the power method and minimization functions so that we can compute the dominant eigenvalues of the data and use it to classify them. We could then identify which set each image belonged to by computing the distances of the vectors' orthogonal projections. Relevant equations include the error for a projection of b onto V (1) and the equation for computing the 2nd-most dominant eigenvalue of a matrix (2) and both of which can be found below.

1. $e = b - p = b - B(B^TB)^{-1}B^Tb$
2. $B = A - \lambda_1 v^{(1)} x^T$

II. Analysis

We can observe that our methods are implemented correctly from our convergence analysis plot of the power method below (Figure 1), as well as by the fact that we can see that our power method converged for each iteration of our method from the MATLAB command window (Figure 2). This can all be seen below in the aforementioned figures.

III. Results and Discussion

Our results as seen in the confusion charts below (Figures 3 and 4) show the predictive accuracy of our algorithm when using two dominant eigenvalues ($r = 2$) and only one ($r = 1$). With only one dominant eigenvalue, we have a total accuracy of $26/30$ or 87 percent. Despite using only one dominant eigenvalue, every diagonal entry has at least one correct prediction and only 3 of them were not perfect. This is exceptional accuracy but this can still be further improved by using two dominant eigenvalues. For two dominant eigenvalues, we see once again that every diagonal entry has at least one correct prediction and $7/10$ have perfect accuracy however this time, none of our diagonal entries misclassified 2 of the images. Our total accuracy was $27/30$ or 90 percent. This is a slight increase in accuracy and highlights the importance of considering more dominant eigenvalues to bolster the predictive accuracy of our classification algorithm as well as the relative strength of the algorithm to begin with.

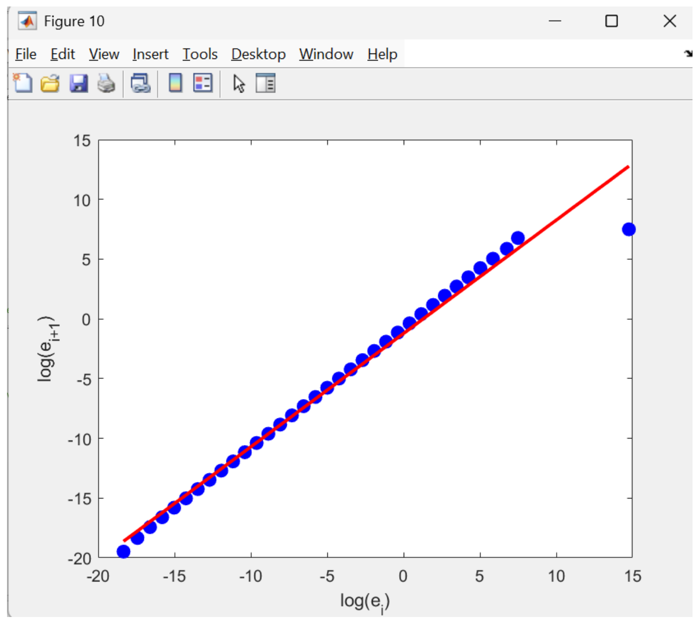

Fig. 1. Convergence analysis plot of the power method for one set of images shows order of 0.919, which is approximately our expected order of 1 meaning linear convergence.

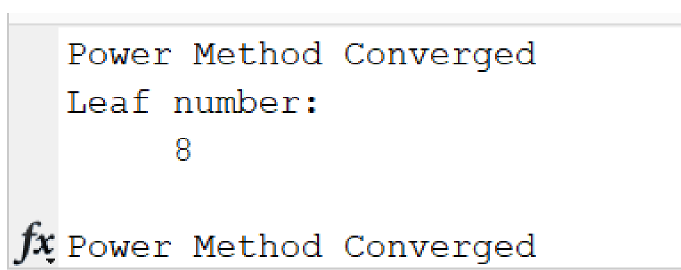

Fig. 2. The MATLAB command window tells us that the power method converged for each iteration.

IV. Future Directions

We could try to extend our methods utilized in this project by trying to create an algorithm that can work on a matrix that isn't diagonalizable. Additionally, we could improve our work by trying to make it more accurate by trying to reduce errors that occur for data with similar matrix structure.

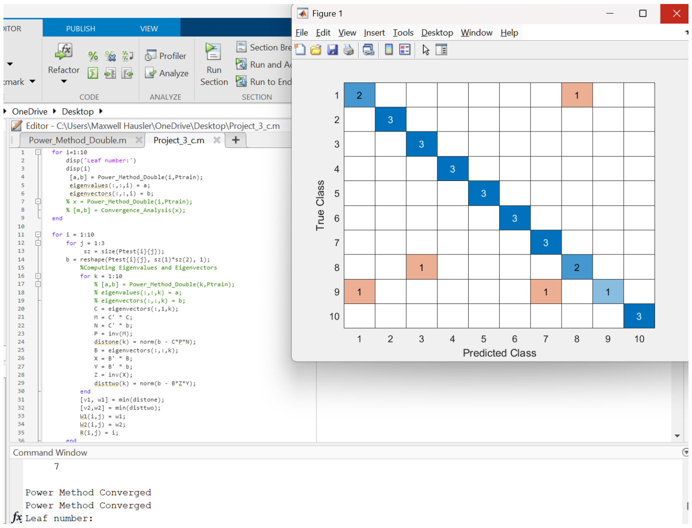

Fig. 3. The confusion chart for one dominant eigenvalue.

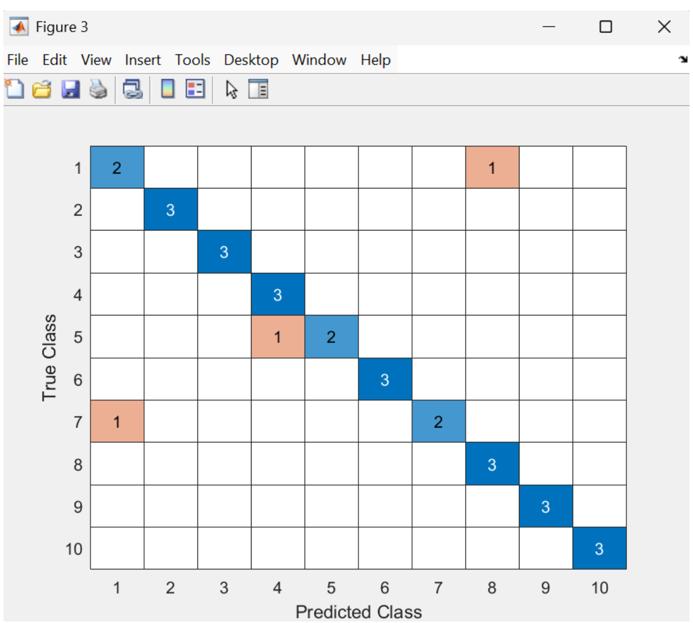

Fig. 4. The confusion chart for two dominant eigenvalues.